

Tasnimul Hoque Rafi

21 Rue André Maginot, Orsay, France

✉ hoquerafi727@gmail.com | 📞 +33605549606

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About Me

I am pursuing my M.Sc. at the Université Paris-Saclay, specializing in Mechatronics, Machine Vision and AI. My research interests lie at the intersection of robotics, computer vision, embedded systems, digital twins, and extended reality. Having developed a solid foundation through a range of relevant projects and research experience, I aim to develop intelligent robotic systems that integrate perception, computation, and autonomous decision-making capabilities.

Education

Postgraduate

Sept 2026 - Sept 2027

[Université Paris-Saclay](#).

M.Sc. in Mechatronics, Machine Vision and AI

Relevant Courses: Advances of Mechatronic Systems, Kinematics and Dynamic Modelling of Mechatronic Systems, Modelling of Parallel Manipulators, Advanced Control of Smart Systems, Robotic System Architecture and Programming, Machine Learning, Advances in Machine Vision.

Undergraduate

Apr 2019 - Jul 2024

[Bangladesh University of Engineering and Technology \(BUET\)](#).

B.Sc. in Electrical and Electronic Engineering (EEE)

Result: **CGPA: /4**

Relevant Courses: Control Systems, Power Electronics, Digital Electronics, Microprocessors and Embedded Systems, Robotics and Automation, Digital Signal Processing, Microwave Engineering, Optical Communication, Radar and Satellite Communication, Wireless Communication.

Higher Secondary Certificate

Jun 2016 - May 2018

[Comilla Victoria Government College](#).

Result: **GPA: 5/5**

Experience

Research Intern

Sept 2025 - March 2026

- **Organisation:** CESI, École d'Ingénieurs | Saint-Étienne-du-Rouvray, Rouen, France.
- **Title:** 3D LiDAR and RGB-D Camera Fusion for Real-Time Robotic Perception. [📺 YouTube](#)
- **Supervisors:** Dr. Fabrice Duval, Dr. Nicolas Ragot, Dr. Argui Imane
- **Summary:** Deployed real time 3D object detection with improved inference time on edge embedded platforms, filtered the estimated depth using histogram based approach combined with instance segmentation, performed camera extrinsic calibration using the OptiTrack Motion Capture System to generate the depth ground truth, extracted point cloud from raw LiDAR data and performed camera-LiDAR extrinsic calibration using least square optimization on the point to point correspondance dataset.

Engineering Intern

Nov 2023

- **Organisation:** Grameenphone Ltd (Telecommunication Service Provider)
- **Summary:** Learned the deployment of BTS(Base Transceiver Station) and types of antennas like GSM and Microwave including the challenges and power management. Explored the core cellular communication technologies like GSM,CDMA,GPRS,EDGE,3G,4G and LTE.

Research

Undergraduate Thesis:

Title: "Virtual Zone Based Traffic Monitoring and Driver Assistance"

Supervisor: Dr. S.M. Mahbubur Rahman, Professor, Department of EEE, BUET.  [Link](#)

Summary: Worked on Computer Vision Based Autonomous Traffic Monitoring on the roads and highways of Bangladesh using deep learning models from video stream. It includes Traffic sign and vehicles classification, Number plate recognition, Speed Estimation, Zone based vehicles counting, Zone based collision avoidance alert, Driver's drowsiness Alert.  [YouTube](#) |  [GitHub](#)

Projects

1. Medicine/Relief Delivery Drone for Flood Affected Region  [YouTube](#)
2. The Aerial Weather Station (Weather Monitoring Drone)  [YouTube](#) |  [GitHub](#)
3. Gesture Controlled Drone Using Computer Vision  [YouTube](#) |  [GitHub](#)
4. Computer Vision Based Gesture Controlled Obstacle Avoiding BOT  [YouTube](#) |  [GitHub](#)
5. Gesture Controlled Robotic Manipulator (2 DoF) Using Computer Vision  [YouTube](#) |  [GitHub](#)
6. Computer Vision Based Speed Control of DC Fan Using Hand Gesture  [YouTube](#) |  [GitHub](#)
7. Computer Vision Based Electric Load Control Using Hand Gesture  [YouTube](#) |  [GitHub](#)
8. Electric Load Control Using Virtual Buttons  [YouTube](#) |  [GitHub](#)
9. Face Recognition/QR Code Based Smart Attendance System  [YouTube](#) |  [GitHub](#)
10. Drowsiness and Yawn Alert for Driver Assistance Using Computer Vision  [YouTube](#) |  [GitHub](#)
11. Voice Controlled Obstacle Avoiding BOT  [YouTube](#) |  [GitHub](#)
12. IoT Based Smart Door Lock Using Password/OTP/Rfid/FaceID/QR Code  [YouTube](#) |  [GitHub](#)
13. IoT Based Patient Monitoring System  [YouTube](#) |  [GitHub](#)
14. IoT Based Weather Station  [YouTube](#) |  [GitHub](#)
15. Data Transmission Using Light (LiFi) with Morse Code Encoding  [YouTube](#) |  [GitHub](#)

Skills

- **Microcontrollers:** Arduino, ESP, STM, FPGA, Raspberry Pi, Jetson Orin Nano
- **Programming Languages:** Python, C/C++, VHDL, Matlab
- **Simulation:** PSpice, AutoCAD, Matlab/Simulink, Quartus, Keil, Proteus, SITL, Gazebo, Unity
- **Frameworks:** ROS2, PyTorch, TensorFlow, Scikit-Learn, OpenCV, DroneKit
- **Document Preparation:** Overleaf (LaTeX), Microsoft Word/PowerPoint/Excel
- **Languages:** English (Professional Fluency), French (Early B1), Italian (A2), Bengali (Native)

Volunteering Activities

- [1] Conducted hands-on workshops on Microcontrollers and UAVs for fellow juniors.
- [2] Taught higher secondary Physics and Math to fellow juniors, reinforcing their fundamental concepts.
- [3] Volunteered in the competitions held by the Robotics Club and helped coordinate cultural events.

References

1. Dr. Samer AlFayad Professor, Université Paris-Saclay Contact: +33 6 95 45 95 24 Email: samer.alfayad@univ-evry.fr  Link	2. Dr. DUVAL Fabrice Internship Supervisor, Professor, CESI Contact: +33 7 61 73 27 19 Email: fduval@cesi.fr  Link
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